

Executive Summary

Restaurant Data Analysis for Business Insights

Objective

The objective of this project is to analyze restaurant data to identify patterns in customer ratings, cuisines, pricing, and location preferences. The analysis aims to generate actionable insights that can support business growth and improve user experience on a restaurant discovery platform.

Methodology

The dataset was cleaned by handling missing values, removing duplicates, standardizing text fields, and converting rating values into numerical format. Exploratory Data Analysis (EDA) was then performed using Python libraries such as Pandas, Matplotlib, and Seaborn.

Several visualizations were developed, including rating distributions, cuisine analysis, location hotspots, price versus rating relationships, correlation heatmaps, and word clouds.

Key Findings

- Most restaurants receive ratings between 3.5 and 4.2.
- Popular restaurant locations have a higher concentration of highly rated establishments.
- Restaurant cost shows only a weak correlation with customer ratings, indicating that expensive restaurants are not necessarily better.
- Certain cuisines consistently achieve higher average ratings than others.
- Restaurants with higher customer engagement (votes) generally receive stronger ratings.

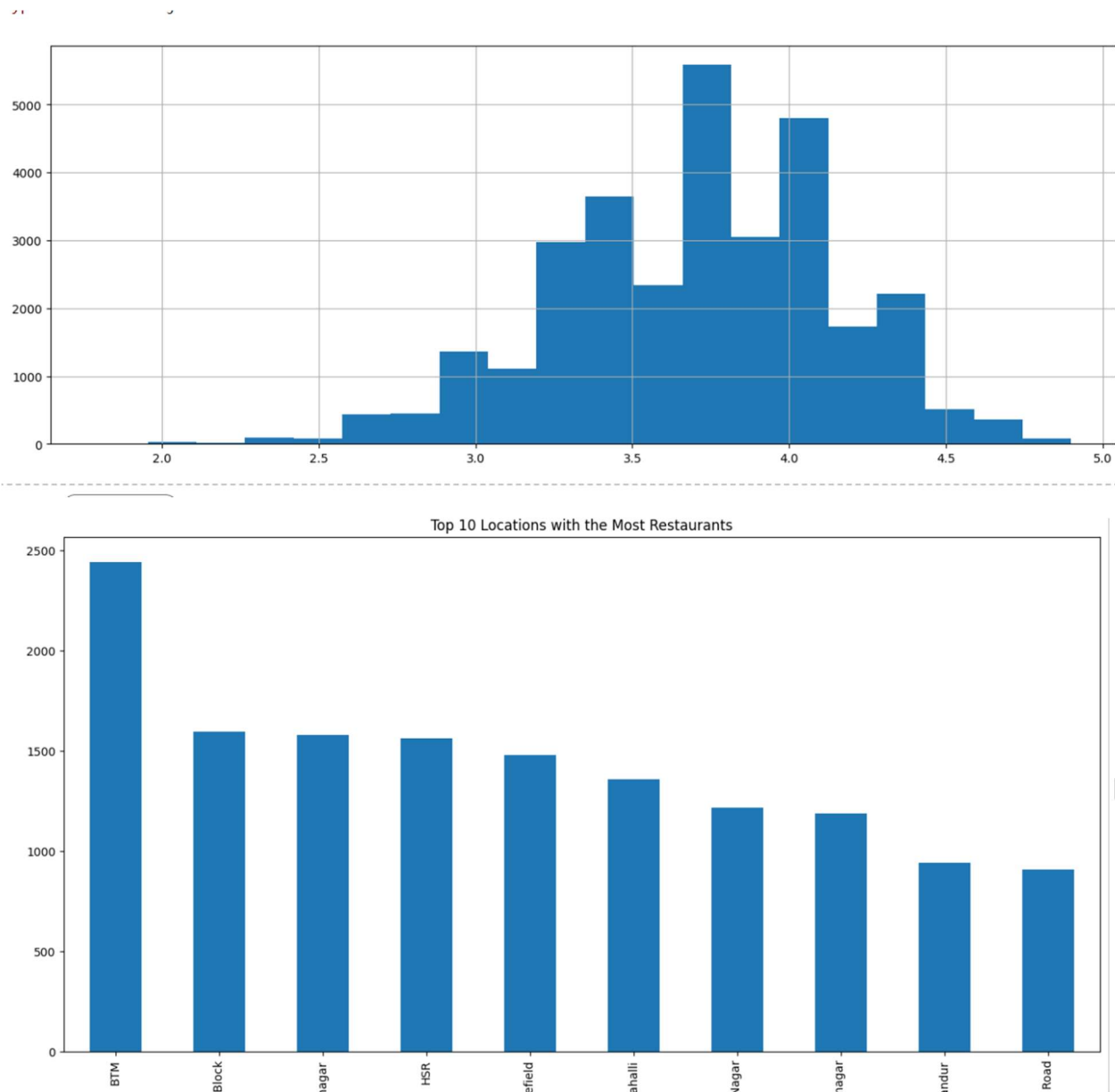
Business Recommendations

1. Partner with highly rated restaurants in emerging locations.
2. Promote top-performing cuisines through curated collections.
3. Build personalized recommendation systems based on cuisine, budget, and location.
4. Publish location-based restaurant guides and food blogs to increase platform engagement.
5. Encourage verified customer reviews through rewards and loyalty programs.

Conclusion

The analysis demonstrates that customer satisfaction depends on multiple factors beyond price, with cuisine quality, popularity, and location playing significant roles. These insights can help platforms optimize partnerships, marketing strategies, and recommendation systems to enhance user satisfaction and business performance.

Screenshots



Top 10 cuisines

